

6. Measurable Functions and Operations on Them

REAL VARIABLES I

Fall 2026

Measures assign sizes to measurable subsets of a space. To integrate a function, we need to know that the subsets selected by its values are also measurable. For a function $f: X \rightarrow Y$, a set $B \subseteq Y$ determines the event

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

When X and Y carry σ -algebras, measurability requires every measurable event in the target to pull back to a measurable event in the domain. This criterion makes functions compatible with the set operations from the previous unit. The payoff is a practical collection of tests and closure results that will let us construct the Lebesgue integral in the next notes.

After studying these notes, you will be able to do the following.

- Verify measurability by testing inverse images of a generating family.
- Construct pullback, product, and trace σ -algebras.
- Apply level-set criteria to real- and extended-real-valued functions.
- Prove that measurable functions remain measurable under pasting, algebraic operations, lattice operations, and pointwise limits when the operations are defined.

6.1 Measurable maps

Definition 6.1.1 (Measurable map). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A function $f: X \rightarrow Y$ is $(\mathcal{A}, \mathcal{B})$ -measurable if

$$f^{-1}(B) \in \mathcal{A} \quad \text{for every } B \in \mathcal{B}.$$

When the σ -algebras are fixed, we say that f is *measurable*.

The definition is expressed through inverse images because inverse images preserve the set operations used in a σ -algebra. For every family $(B_i)_{i \in I}$ of subsets of Y ,

$$\begin{aligned} f^{-1}(Y) &= X, \\ f^{-1}(Y \setminus B) &= X \setminus f^{-1}(B), \\ f^{-1}\left(\bigcup_{i \in I} B_i\right) &= \bigcup_{i \in I} f^{-1}(B_i), \\ f^{-1}\left(\bigcap_{i \in I} B_i\right) &= \bigcap_{i \in I} f^{-1}(B_i). \end{aligned}$$

These identities hold for arbitrary unions and intersections, not only for countable ones.

Proposition 6.1.2 (Composition of measurable maps). Let

$$(X, \mathcal{A}) \xrightarrow{f} (Y, \mathcal{B}) \xrightarrow{g} (Z, \mathcal{C})$$

be measurable maps. Then $g \circ f: (X, \mathcal{A}) \rightarrow (Z, \mathcal{C})$ is measurable.

Proof. (proves 6.1.2)

The inverse image of a set under a composition is an iterated inverse image. For $C \in \mathcal{C}$, measurability of g gives $g^{-1}(C) \in \mathcal{B}$. Measurability of f then gives

$$(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C)) \in \mathcal{A}.$$

□

The inverse images of all measurable target sets themselves form a σ -algebra.

Definition 6.1.3 (Pullback σ -algebra). Let $f: X \rightarrow Y$ be a function and let $\mathcal{B}$ be a σ -algebra on Y . The *pullback σ -algebra* is

$$f^{-1}(\mathcal{B}) = \{f^{-1}(B) : B \in \mathcal{B}\}.$$

Proposition 6.1.4 (Minimality of the pullback). The collection $f^{-1}(\mathcal{B})$ is a σ -algebra on X . It is the smallest σ -algebra on X for which $f: X \rightarrow (Y, \mathcal{B})$ is measurable.

Proof. (proves 6.1.4)

The inverse-image identities verify the σ -algebra axioms, and the definition of measurability then gives minimality. The inverse-image identities give

$$X = f^{-1}(Y) \in f^{-1}(\mathcal{B}),$$

and show that $f^{-1}(\mathcal{B})$ is closed under complements and countable unions. Thus it is a σ -algebra.

If $\mathcal{A}$ is any σ -algebra on X for which f is measurable, then $f^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}$. Hence $f^{-1}(\mathcal{B}) \subseteq \mathcal{A}$, which proves the minimality assertion. □

6.1.1 Checking a generating family

The definition quantifies over every member of the target σ -algebra. It is enough to check a family that generates that σ -algebra.

Proposition 6.1.5 (Generator criterion for measurability). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces, and suppose that $\mathcal{G} \subseteq 2^Y$ satisfies $\mathcal{B} = \sigma(\mathcal{G})$. A function $f: X \rightarrow Y$ is measurable if and only if

$$f^{-1}(G) \in \mathcal{A} \quad \text{for every } G \in \mathcal{G}.$$

Proof. (proves 6.1.5)

The reverse implication collects all target sets with measurable inverse image into a σ -algebra. The forward implication follows from $\mathcal{G} \subseteq \mathcal{B}$. For the reverse implication, define

$$\mathcal{H} = \{E \subseteq Y : f^{-1}(E) \in \mathcal{A}\}.$$

The inverse-image identities show that $\mathcal{H}$ is a σ -algebra on Y . The hypothesis gives $\mathcal{G} \subseteq \mathcal{H}$, and therefore

$$\mathcal{B} = \sigma(\mathcal{G}) \subseteq \mathcal{H}.$$

Thus $f^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}$. □

Example 6.1.6 (Constant functions). Every constant function $f: (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ is measurable. Indeed, if $f(x) = y_0$ for every $x \in X$, then

$$f^{-1}(B) = \begin{cases} X, & y_0 \in B, \\ \emptyset, & y_0 \notin B. \end{cases}$$

Both possible inverse images belong to $\mathcal{A}$.

Example 6.1.7 (Indicator functions). For $E \subseteq X$, define the indicator function

$$\mathbb{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

Give $\{0, 1\}$ its full σ -algebra. Then $\mathbb{1}_E: (X, \mathcal{A}) \rightarrow \{0, 1\}$ is measurable if and only if $E \in \mathcal{A}$. The nontrivial implication follows from

$$E = \mathbb{1}_E^{-1}(\{1\}).$$

For the reverse implication, every inverse image under $\mathbb{1}_E$ is one of $\emptyset$, E , $X \setminus E$, or X . The same equivalence holds when $\mathbb{1}_E$ is viewed as an $\mathbb{R}$ -valued function: the inverse image of a Borel set depends only on its intersection with $\{0, 1\}$.

6.1.2 Product measurable spaces

Definition 6.1.8 (Product σ -algebra). For measurable spaces $(Y, \mathcal{B})$ and $(Z, \mathcal{C})$, their product σ -algebra is

$$\mathcal{B} \otimes \mathcal{C} = \sigma(\{B \times C : B \in \mathcal{B}, C \in \mathcal{C}\}).$$

Example 6.1.9 (Coordinate projections). The coordinate projections

$$\pi_Y: (Y \times Z, \mathcal{B} \otimes \mathcal{C}) \rightarrow (Y, \mathcal{B}), \quad \pi_Z: (Y \times Z, \mathcal{B} \otimes \mathcal{C}) \rightarrow (Z, \mathcal{C})$$

are measurable. For $B \in \mathcal{B}$ and $C \in \mathcal{C}$,

$$\pi_Y^{-1}(B) = B \times Z, \quad \pi_Z^{-1}(C) = Y \times C,$$

and both sets belong to $\mathcal{B} \otimes \mathcal{C}$.

Proposition 6.1.10 (Measurability of a paired map). Let $f: (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ and $g: (X, \mathcal{A}) \rightarrow (Z, \mathcal{C})$, and define

$$(f, g)(x) = (f(x), g(x)).$$

Then $(f, g): (X, \mathcal{A}) \rightarrow (Y \times Z, \mathcal{B} \otimes \mathcal{C})$ is measurable if and only if f and g are measurable.

Proof. (proves 6.1.10)

The two directions use the rectangle generators and the coordinate projections, respectively.

Step 1: test the paired map on measurable rectangles.

Suppose first that f and g are measurable. For every measurable rectangle $B \times C$,

$$(f, g)^{-1}(B \times C) = f^{-1}(B) \cap g^{-1}(C) \in \mathcal{A}.$$

The measurable rectangles generate $\mathcal{B} \otimes \mathcal{C}$, so [Proposition 6.1.5](#) applies.

Step 2: recover the coordinate maps by projection.

Conversely, if (f, g) is measurable, then

$$f = \pi_Y \circ (f, g), \quad g = \pi_Z \circ (f, g).$$

The coordinate projections are measurable by [Example 6.1.9](#); hence [Proposition 6.1.2](#) gives the conclusion. $\square$

For Euclidean spaces, $\mathcal{B}(\mathbb{R}^m) \otimes \mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R}^{m+n})$. One inclusion follows because rectangles with open factors are Borel. For the other inclusion, open rectangles with rational endpoints form a countable basis for $\mathbb{R}^{m+n}$, so every open set is a countable union of such rectangles. Consequently, a map $F: X \rightarrow \mathbb{R}^d$ is measurable if and only if each of its coordinate functions is measurable.

6.1.3 Borel and Lebesgue measurability

For a topological space Y , write $\mathcal{B}(Y)$ for the σ -algebra generated by the open subsets of Y .

Theorem 6.1.11 (Continuous maps are Borel measurable). Let X and Y be topological spaces. Every continuous map

$$f: (X, \mathcal{B}(X)) \rightarrow (Y, \mathcal{B}(Y))$$

is measurable. $\square$

Proof. (proves 6.1.11)

The generator criterion reduces the claim to inverse images of open sets. The open subsets of Y generate $\mathcal{B}(Y)$. If $V \subseteq Y$ is open, continuity gives that $f^{-1}(V)$ is open in X , and hence $f^{-1}(V) \in \mathcal{B}(X)$. Apply [Proposition 6.1.5](#). $\square$

Definition 6.1.12 (Borel and Lebesgue measurable functions). A function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is

1. *Borel measurable* if it is $(\mathcal{B}(\mathbb{R}^d), \mathcal{B}(\mathbb{R}))$ -measurable;
2. *Lebesgue measurable* if it is $(\mathcal{L}(\mathbb{R}^d), \mathcal{B}(\mathbb{R}))$ -measurable, where $\mathcal{L}(\mathbb{R}^d)$ is the Lebesgue σ -algebra.

Since $\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d)$, every Borel measurable real-valued function on $\mathbb{R}^d$ is Lebesgue measurable. The converse does not hold.

Example 6.1.13 (A Lebesgue measurable function that is not Borel measurable). Let $C \subseteq [0, 1]$ be the Cantor set. It is a Borel set of Lebesgue measure zero. Completeness of Lebesgue measure implies that every subset of C is Lebesgue measurable. The set C has the cardinality of $\mathbb{R}$, and therefore its power set has cardinality strictly larger than $|\mathbb{R}|$. There are only $|\mathbb{R}|$ Borel subsets of $\mathbb{R}$, since each Borel set has a countable code built from a countable topological basis. Consequently, some subset $E \subseteq C$ is not Borel. By [Example 6.1.7](#), the function $\mathbb{1}_E: (\mathbb{R}, \mathcal{L}(\mathbb{R})) \rightarrow \mathbb{R}$ is measurable. It is not Borel measurable because

$$E = \mathbb{1}_E^{-1}((1/2, 3/2))$$

is not a Borel set. $\square$

6.2 Level-set criteria

For a real-valued function, measurability can be checked using only one family of rays. This turns questions about operations on functions into set identities involving inequalities.

Proposition 6.2.1 (Level-set criterion for real-valued functions). Let $(X, \mathcal{A})$ be a measurable space and let $f: X \rightarrow \mathbb{R}$. The following conditions are equivalent:

1. f is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ -measurable;
2. $\{x \in X : f(x) > a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
3. $\{x \in X : f(x) \geq a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
4. $\{x \in X : f(x) < a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
5. $\{x \in X : f(x) \leq a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.

In condition 2, it is enough to use $a \in \mathbb{Q}$. The same rational restriction can be made in conditions 3–5.

Proof. (proves 6.2.1)

The proof first identifies one countable generating family of rays and then passes between the four inequality conventions.

Step 1: generate the Borel sets from rational upper rays.

The rays (q, ∞) , $q \in \mathbb{Q}$, generate $\mathcal{B}(\mathbb{R})$. To verify this assertion, first observe that

$$(a, \infty) = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} (q, \infty).$$

Also,

$$[b, \infty) = \bigcap_{\substack{q \in \mathbb{Q} \\ q < b}} (q, \infty), \quad (-\infty, b) = \mathbb{R} \setminus [b, \infty).$$

Thus every open interval belongs to the σ -algebra generated by these rational rays. Every open subset of $\mathbb{R}$ is a countable union of open intervals with rational endpoints, so the rational rays generate all of $\mathcal{B}(\mathbb{R})$. The equivalence of conditions 1 and 2 now follows from [Proposition 6.1.5](#).

Step 2: pass between strict and non-strict level sets.

The remaining choices of inequality are equivalent by complements and countable approximations. For example,

$$\begin{aligned} \{f \geq a\} &= \bigcap_{n=1}^{\infty} \{f > a - 2^{-n}\}, \\ \{f < a\} &= X \setminus \{f \geq a\}, \\ \{f \leq a\} &= X \setminus \{f > a\}, \\ \{f > a\} &= \bigcup_{n=1}^{\infty} \{f \geq a + 2^{-n}\}. \end{aligned}$$

The rational versions follow from the same generator argument. □

The distinction between strict and non-strict inequalities matters when a set identity is formed. For example,

if $h(x) = \sup_{n \in \mathbb{N}} f_n(x)$, then

$$\{h \leq a\} = \bigcap_{n \in \mathbb{N}} \{f_n \leq a\}, \quad \{h > a\} = \bigcup_{n \in \mathbb{N}} \{f_n > a\}.$$

In contrast, the condition $f_n(x) < a$ for every n does not force $\sup_n f_n(x) < a$.

6.3 Operations on real-valued measurable functions

Proposition 6.3.1 (Measurable pasting). Let $E \in \mathcal{A}$. Suppose that $f: E \rightarrow (Y, \mathcal{B})$ and $g: X \setminus E \rightarrow (Y, \mathcal{B})$ are measurable when their domains carry the corresponding trace σ -algebras. Define

$$h(x) = \begin{cases} f(x), & x \in E, \\ g(x), & x \notin E. \end{cases}$$

Then h is measurable.

Proof. (proves 6.3.1)

Each inverse image splits into one measurable piece from each trace space. For $B \in \mathcal{B}$, both $f^{-1}(B)$ and $g^{-1}(B)$ belong to $\mathcal{A}$: each belongs to the relevant trace σ -algebra, and $E \in \mathcal{A}$. Therefore

$$h^{-1}(B) = f^{-1}(B) \cup g^{-1}(B) \in \mathcal{A}.$$

□

Theorem 6.3.2 (Algebra of real-valued measurable functions). Let $f, g: (X, \mathcal{A}) \rightarrow \mathbb{R}$ be measurable and let $\alpha \in \mathbb{R}$. Then the following functions are measurable:

1. $f + g$, $f - g$, fg , and αf ;
2. $\max\{f, g\}$, $\min\{f, g\}$, and $|f|$;
3. the positive and negative parts

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\};$$

4. f/g on the measurable subspace $\{x \in X : g(x) \neq 0\}$.

If f/g is assigned any fixed real value on $\{g = 0\}$, the resulting function on X is also measurable.

Proof. (proves 6.3.2)

The proof establishes addition by a countable level-set identity, derives the algebra and lattice operations, and treats division on its measurable domain.

Step 1: prove measurability of addition from rational level sets.

We first prove measurability of $f + g$ directly from level sets. For $a \in \mathbb{R}$, density of $\mathbb{Q}$ gives the identity

$$\{f + g < a\} = \bigcup_{\substack{(q,r) \in \mathbb{Q}^2 \\ q+r < a}} (\{f < q\} \cap \{g < r\}).$$

If $f(x) < q$, $g(x) < r$, and $q + r < a$, then $f(x) + g(x) < a$. Conversely, if $f(x) + g(x) < a$, choose positive

$\varepsilon < a - f(x) - g(x)$. Density of $\mathbb{Q}$ supplies $q, r \in \mathbb{Q}$ such that

$$f(x) < q < f(x) + \varepsilon/2, \quad g(x) < r < g(x) + \varepsilon/2.$$

Then $q + r < a$. The displayed union is countable, and each of its terms is in $\mathcal{A}$, so $f + g$ is measurable by [Proposition 6.2.1](#).

Step 2: derive the remaining algebra and lattice operations.

The function $t \mapsto -t$ is continuous, hence Borel measurable by [Theorem 6.1.11](#). Composition shows that $-g$ is measurable, and therefore $f - g = f + (-g)$ is measurable. The same argument with $t \mapsto \alpha t$, $t \mapsto t^2$, and $t \mapsto |t|$ proves measurability of αf , f^2 , and $|f|$. The identity

$$fg = \frac{(f + g)^2 - f^2 - g^2}{2}$$

then proves measurability of the product.

The formulas

$$\max\{f, g\} = \frac{f + g + |f - g|}{2}, \quad \min\{f, g\} = \frac{f + g - |f - g|}{2}$$

give the maximum and minimum. Taking $g = 0$ and using measurability of $-f$ gives f^+ and f^- . In particular,

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Step 3: handle division on its natural domain.

The set $D = \{g \neq 0\}$ is measurable by [Proposition 6.2.1](#). The restrictions of f and g to D , where D has the trace σ -algebra, are measurable, and $g(D) \subseteq \mathbb{R} \setminus \{0\}$. Since $t \mapsto 1/t$ is continuous on $\mathbb{R} \setminus \{0\}$, the function $1/g$ is measurable on D , and so is $f/g = f(1/g)$. Finally, $X \setminus D = \{g = 0\}$ is measurable. Pasting f/g on D with a constant function on $X \setminus D$ proves the last assertion by [Proposition 6.3.1](#). $\square$

The proof of addition uses rational parameters because a σ -algebra is closed under countable unions, not arbitrary unions. The pairing result gives another proof: $(f, g): X \rightarrow \mathbb{R}^2$ is measurable and addition, multiplication, maximum, and minimum are continuous maps from $\mathbb{R}^2$ to $\mathbb{R}$.

6.4 Extended-real-valued functions and pointwise limits

Let

$$\bar{\mathbb{R}} = [-\infty, +\infty]$$

with its order topology and Borel σ -algebra $\mathcal{B}(\bar{\mathbb{R}})$. The order topology has neighborhoods of $+\infty$ of the form $(a, +\infty]$ and neighborhoods of $-\infty$ of the form $[-\infty, a)$.

Proposition 6.4.1 (Extended-real level-set criterion). Let $f: X \rightarrow \bar{\mathbb{R}}$. Then f is $(\mathcal{A}, \mathcal{B}(\bar{\mathbb{R}}))$ -measurable if and only if

$$\{x \in X : f(x) > a\} \in \mathcal{A} \quad \text{for every } a \in \mathbb{R}.$$

One may instead use any one of the other three inequalities $f \geq a$, $f < a$, or $f \leq a$. It is enough to use rational a .

Proof. (proves [6.4.1](#))

The proof augments the real-valued ray argument with the two infinite endpoints. The rays $(a, +\infty]$, $a \in \mathbb{R}$, generate $\mathcal{B}(\bar{\mathbb{R}})$. They generate the bounded open intervals by the same complement and countable-

approximation identities used in [Proposition 6.2.1](#). They also generate the two endpoints because

$$\{+\infty\} = \bigcap_{n=1}^{\infty} (n, +\infty], \quad \{-\infty\} = \bar{\mathbb{R}} \setminus \bigcup_{n=1}^{\infty} (-n, +\infty].$$

The generator criterion proves the first equivalence. Complements and the identities from [Proposition 6.2.1](#) give the other inequality choices and the rational restriction. $\square$

The sets on which an extended-real-valued measurable function takes an infinite value are measurable. In fact,

$$\{f = +\infty\} = \bigcap_{n=1}^{\infty} \{f > n\}, \quad \{f = -\infty\} = \bigcap_{n=1}^{\infty} \{f < -n\}.$$

Theorem 6.4.2 (Countable lattice operations and pointwise limits). Let $(f_n : X \rightarrow \bar{\mathbb{R}})_{n \in \mathbb{N}}$ be a sequence of measurable functions. Then the following extended-real-valued functions are measurable:

1. $\sup_{n \in \mathbb{N}} f_n$ and $\inf_{n \in \mathbb{N}} f_n$;
2. $\limsup_{n \rightarrow \infty} f_n$ and $\liminf_{n \rightarrow \infty} f_n$.

Consequently, if $f_n(x) \rightarrow f(x)$ in $\bar{\mathbb{R}}$ for every $x \in X$, then f is measurable.

Proof. (proves 6.4.2)

The proof handles suprema and infima by level sets, builds limsup and liminf from tail operations, and then identifies an existing pointwise limit.

Step 1: prove measurability of countable suprema and infima.

Define $s(x) = \sup_{n \in \mathbb{N}} f_n(x)$. For every $a \in \mathbb{R}$,

$$\{s > a\} = \bigcup_{n \in \mathbb{N}} \{f_n > a\} \in \mathcal{A}.$$

Thus s is measurable by [Proposition 6.4.1](#). If $i(x) = \inf_{n \in \mathbb{N}} f_n(x)$, then

$$\{i < a\} = \bigcup_{n \in \mathbb{N}} \{f_n < a\} \in \mathcal{A}.$$

The version of the level-set criterion using strict sublevel sets proves that i is measurable.

Step 2: form limsup and liminf from tail operations.

For every $N \in \mathbb{N}$, the tail supremum and tail infimum

$$S_N = \sup_{n \geq N} f_n, \quad I_N = \inf_{n \geq N} f_n$$

are measurable by the first part. The identities

$$\limsup_{n \rightarrow \infty} f_n = \inf_{N \in \mathbb{N}} S_N, \quad \liminf_{n \rightarrow \infty} f_n = \sup_{N \in \mathbb{N}} I_N$$

then prove the second part.

Step 3: identify a pointwise limit.

If $f_n(x) \rightarrow f(x)$, then at every $x \in X$,

$$f(x) = \liminf_{n \rightarrow \infty} f_n(x) = \limsup_{n \rightarrow \infty} f_n(x).$$

Hence f is measurable. $\square$

Even if every f_n is real-valued, a countable supremum, infimum, limsup, or liminf can take an infinite value. The codomain $\mathbb{R}$ makes each of these operations defined without an additional boundedness hypothesis.

Example 6.4.3 (A discontinuous pointwise limit). On $[0, 1]$, let $f_n(x) = x^n$. Every f_n is continuous and hence Borel measurable. Pointwise,

$$f_n(x) \longrightarrow \mathbb{1}_{\{1\}}(x).$$

The limit is discontinuous at 1, but it is measurable by [Theorem 6.4.2](#). Thus pointwise limits preserve measurability even though they need not preserve continuity.

Warning (Uncountable suprema). Countability in [Theorem 6.4.2](#) cannot be removed. Let $V \subseteq \mathbb{R}$ be a non-Borel set. For every $v \in V$, the function $\mathbb{1}_{\{v\}}$ is Borel measurable, but

$$\sup_{v \in V} \mathbb{1}_{\{v\}} = \mathbb{1}_V$$

is not Borel measurable. The superlevel set of the supremum is an uncountable union of measurable singletons, and a σ -algebra need not be closed under such a union.

6.4.1 Arithmetic at infinity

Arithmetic on $\bar{\mathbb{R}}$ requires conventions that are absent for real-valued functions. Negation is defined by

$$-(+\infty) = -\infty, \quad -(-\infty) = +\infty.$$

For addition, $a + (+\infty) = +\infty$ and $a + (-\infty) = -\infty$ for finite a . The expressions $(+\infty) + (-\infty)$ and $(-\infty) + (+\infty)$ remain undefined. Thus a pointwise sum $f + g$ is a function only when neither indeterminate pair occurs.

For multiplication, these notes use

$$0(\pm\infty) = 0.$$

If $a \in \mathbb{R} \setminus \{0\}$, the value of $a(\pm\infty)$ is determined by the usual sign rule. Products of two infinite values are determined by the same rule. This convention is suited to non-negative integration, where an integrand that vanishes contributes zero even on a set on which a second factor is infinite.

Proposition 6.4.4 (Measurability of extended-real arithmetic). Let $f, g: (X, \mathcal{A}) \rightarrow \bar{\mathbb{R}}$ be measurable.

1. The functions $-f$, $|f|$, $\max\{f, g\}$, and $\min\{f, g\}$ are measurable.
2. If $f + g$ is defined at every point, then $f + g$ is measurable. The same conclusion holds for $f - g$ whenever it is defined at every point.
3. Under the convention $0(\pm\infty) = 0$, the product fg is measurable.

Proof. (proves [6.4.4](#))

The proof treats the continuous lattice operations first and then obtains sums and products as pointwise limits of finite truncations.

Step 1: handle negation and lattice operations.

The map $t \mapsto -t$ is continuous on $\bar{\mathbb{R}}$, so composition proves measurability of $-f$. For $a \in \mathbb{R}$,

$$\begin{aligned} \{\max\{f, g\} > a\} &= \{f > a\} \cup \{g > a\}, \\ \{\min\{f, g\} > a\} &= \{f > a\} \cap \{g > a\}. \end{aligned}$$

The extended-real level-set criterion gives measurability of the maximum and minimum. Since $|f| =$

$\max\{f, -f\}$, it also gives measurability of $|f|$.

Step 2: truncate to real-valued measurable functions.

For $k \geq 1$, define the truncation

$$T_k(t) = \max\{-k, \min\{t, k\}\}, \quad t \in \bar{\mathbb{R}}.$$

Here $T_k(-\infty) = -k$ and $T_k(+\infty) = k$. The function $T_k: \bar{\mathbb{R}} \rightarrow \mathbb{R}$ is continuous, so $T_k \circ f$ and $T_k \circ g$ are real-valued measurable functions. By [Theorem 6.3.2](#),

$$(T_k \circ f) + (T_k \circ g) \quad \text{and} \quad (T_k \circ f)(T_k \circ g)$$

are measurable.

Step 3: pass the truncated operations to pointwise limits.

If the pointwise sum $f + g$ contains no indeterminate pair, then

$$(T_k \circ f)(x) + (T_k \circ g)(x) \longrightarrow f(x) + g(x)$$

for every $x \in X$. Under the multiplication convention in the statement,

$$(T_k \circ f)(x)(T_k \circ g)(x) \longrightarrow f(x)g(x)$$

for every $x \in X$, including when one factor is zero and the other is infinite. The pointwise-limit conclusion of [Theorem 6.4.2](#) proves the claims about sums and products. Apply the sum result to $f + (-g)$ for the difference. □

No universal convention is imposed for quotients involving zero or for indeterminate sums. In an application, first specify the exceptional set and the values assigned there. If the exceptional set is measurable, [Proposition 6.3.1](#) reduces measurability to the function on its complement.